1 Measures and Their Constructions

Definition 1. Given a sequence of set $\{E_n\}$:

$$\limsup E_n = \bigcap_{k=1}^{\infty} \bigcup_{n=k}^{\infty} E_n, \quad \liminf E_n = \bigcup_{k=1}^{\infty} \bigcap_{n=k}^{\infty} E_n.$$

Theorem 1 (Disjoint Trick)**.** *Given countable collection $\{F_j\} \subset \mathcal{M}$ such that $\cup_{j=1}^{\infty} F_j$ covers X. Now, define*

$$G_j = F_j \setminus \left[\bigcup_{n=1}^{j-1} F_n\right] = F_j \cap \left(\bigcup_{n=1}^{j-1} F_n\right)^c,$$

then $\{G_j\} \subset \mathcal{M}$ is a disjoint countable collection. It also covers X as $\bigcup F_j = \bigcup G_j$. Also, we have $G_j \subset F_j$ for all j.

Definition 2. Given a σ-algebra $\mathcal{M}$, $\mu : \mathcal{M} \to [0, \infty]$ is a **measure** if it satisfies $\mu(\emptyset) = 0$ and countable additivity.

1. μ is **finite** if $\mu(X) < \infty$,
2. μ is **semifinite** if for all $E \in \mathcal{M}$ such that $\mu(E) = \infty$, there is $F \subset E$ with $0 < \mu(F) < \infty$.
3. μ is **σ-finite** if THERE EXISTS collection $\{E_k\}_{k=1}^{\infty} \supset X$ such that $\mu(E_k) < \infty$ for all k.

Theorem 2. *Let $(X, \mathcal{M}, \mu)$ be a measure space.*

1. *(Monotonicity) If $E, F \in \mathcal{M}$ and $E \subset F$, then $\mu(E) \leq \mu(F)$.*
2. *(Subadditivity) If $\{E_j\}_1^{\infty} \subset \mathcal{M}$, then $\mu\left(\bigcup_1^{\infty} E_j\right) \leq \sum_1^{\infty} \mu\left(E_j\right)$.*
3. *(Continuity from below) If $\{E_j\}_1^{\infty} \subset \mathcal{M}$ and $E_1 \subset E_2 \subset \cdots$, then $\mu\left(\cup_1^{\infty} E_j\right) = \lim_{j\to\infty} \mu\left(E_j\right)$*
4. *(Continuity from above) If $\{E_j\}_1^{\infty} \subset \mathcal{M}, E_1 \supset E_2 \supset \cdots$, and $\mu\left(E_n\right) < \infty$ for some n, then $\mu\left(\cap_1^{\infty} E_j\right) = \lim_{j\to\infty} \mu\left(E_j\right)$*

Theorem 3. *Let $\mathcal{E} \subset \mathcal{P}(X)$ and $\rho : \mathcal{E} \to [0, \infty]$ be such that $\varnothing \in \mathcal{E}$, $X \in \mathcal{E}$, and $\rho(\varnothing) = 0$. For any $A \subset X$, let*

$$\mu^*(A) = \inf\left\{\sum_1^{\infty} \rho\left(E_j\right) : E_j \in \mathcal{E} \text{ and } A \subset \bigcup_1^{\infty} E_j\right\}.$$

*Then μ^* is an **outer measure**, that is, $\mu(\emptyset) = 0, \mu^*(\bigcup_1^{\infty} A_j) \leq \sum_1^{\infty} \mu^*(A_j)$.*

Definition 3. The collection of **μ^*-measurable sets** $\mathcal{M}$ is defined by:

$$\mathcal{M} := \{A : \mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c) \text{ for all } E \subseteq X\}$$

Definition 4. μ_0 is a **premeasure** defined on an **algebra** (closed under finite union and complements) if μ_0 satisfies the requirement of a measure except the domain.

Theorem 4 (Validity of Carathédory Criterion)**.**

1. *$(X, \mathcal{M}, \mu^*|_{\mathcal{M}})$ is a measure space.*
2. *If $\rho = \mu_0$ is a premeasure defined on an algebra $\mathcal{A}$, then every set in $\mathcal{A}$ is μ^*-measurable. Moreover, $\mu^*|_{\mathcal{A}} = \mu_0$.*
3. *(Essentially unique extension of μ_0) If μ, υ extend the measure μ_0 in $\mathcal{A}$:*
   - *(a) Then μ, υ can be extended to $\mathcal{M}$, the measurable sets.*
   - *(b) $\upsilon(E) \leq \mu(E)$ for all $E \in \mathcal{M}$.*
   - *(c) If μ is σ-finite measure, then $\mu = \upsilon$, i.e. μ is unique.*

2 Borel and Lebesgue Measure

Definition 5. An **h-interval** is a half open interval $(a, b]$ or (a, ∞). And the **Borel set** $\mathcal{B}_{\mathbb{R}}$ is the σ-algebra of collection of open sets in $\mathbb{R}$.

Theorem 5. *Let $F : \mathbb{R} \to \mathbb{R}$ be increasing and right continuous. If $(a_j, b_j]$ $(j = 1, \ldots, n)$ are disjoint* h*-intervals, let*

$$\mu\left(\bigcup_1^n (a_j, b_j]\right) = \sum_1^n \left[F(b_j) - F(a_j)\right],$$

and let $\mu(\varnothing) = 0$. Then μ defines a premeasure on the algebra of h-intervals.

After checking μ is indeed σ-finite, follows from a neat trick $\mathbb{R} = \bigcup_{j\in\mathbb{Z}}(j, j+1]$, it follows that there is an "almost 1-to-1" correspondence with μ and the associated function F:

Theorem 6.

1. *Given F, μ is unique (by theorem 4 above) and defines a measure on $\mathcal{B}_{\mathbb{R}}$.*
2. *Given $\mu : \mathcal{B}_{\mathbb{R}} \to [0, \infty]$ that is finite on all bounded Borel set then we can find F by* $F(x) = \begin{cases} \mu((0, x]) & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -\mu((x, 0]) & \text{if } x < 0 \end{cases}$
3. *Given two functions F, G, and define their measure associated with the functions, the measure is the same function if $F - G =$ constant.*

We also make a remark on complete measures:

Definition 6. A measure $\mu : \mathcal{M} \to [0, \infty]$ is **complete** if all the set E, such that $\mu(E) = 0$, we have $F \in \mathcal{M}$ for all $F \subset E$.

Theorem 7. *Every measure have a unique completion (which is also a measure). Suppose $(X, \mathcal{M}, \mu)$ is a measure space. Let $\mathcal{N} = \{N \in \mathcal{M} : \mu(N) = 0\}$ and $\overline{\mathcal{M}} = \{E \cup F : E \in \mathcal{M} \text{ and } F \subset N \text{ for some } N \in \mathcal{N}\}$. Then $\overline{\mathcal{M}}$ is a σ-algebra, and there is a unique extension $\bar{\mu}$ of μ to a complete measure on $\overline{\mathcal{M}}$.*

So given a function F, extend the construction of μ associated to F to find a complete measure, this is the **Lebesgue-Stieltjes Measure** associated to F, we denote it by μ_F.

Theorem 8 (Calculating $\mu(E)$)**.**

$$\mu(E) = \inf\left\{\sum_1^\infty \mu((a_j, b_j)) : E \subset \bigcup_1^\infty (a_j, b_j)\right\} = \inf\{\mu(U) : U \text{ open}, U \supset E\} = \inf\{\mu(K) : K \text{ compact}, K \subset E\}$$

Take $F(x) = x$ and we have the **Lebesgue measure**. It is translational invariant and linear under scaling. There are still ugly sets such as the Cantor set:

Theorem 9. *Let C be the Cantor set, the set $(0, 1)$, divide the interval into 3 parts, remove the middle, and repeat for the remaining 2^n intervals.*
Then C is

- *measurable (since it is a countable union of closed sets),*
- *compact,*
- *nowhere dense (closure of C - in this case C, has empty interior),*
- *C has no isolated point,*
- *$m(C) = 0$,*
- *and C is an uncountable set.*

3 Measurable Functions

Definition 7. Given two measure space $(X, \mathcal{M}), (Y, \mathcal{N})$ and a function between $f : X \to Y$ is $(\mathcal{M}, \mathcal{N})$**-measurable** if $f^{-1}(F) \in \mathcal{M}$ for all $F \in \mathcal{N}$.
$f : X \to \mathbb{R}$, or $X \to \mathbb{C}$ is $(\mathcal{M})$-**measurable** if f is $(\mathcal{M}, \mathcal{B}_{\mathbb{R}})$-measurable or $(\mathcal{M}, \mathcal{B}_{\mathbb{R}})$-measurable.

Theorem 10. *A function $f : X \to \mathbb{R}$ is measurable iff $f^{-1}(\mathcal{E}) \in \mathcal{M}$, where $\mathcal{E}$ generates $\mathcal{B}_{\mathbb{R}}$, so it suffices to check for preimage of (countable) open intervals.*

Definition 8. Define the extended real number $\overline{\mathbb{R}} = \mathbb{R} \cup \{\pm\infty\}$, then $\mathcal{B}_{\overline{\mathbb{R}}} := \{$ Sets where when intersected with $\mathbb{R}$, gives a set in $\mathcal{B}_{\mathbb{R}}\}$.

So this begs the question: How will we identify measurable functions?

Theorem 11. *If $\{f_j\}$ is a sequence of measurable functions, and $f : X \to \overline{R}$ is a function:*

1. $\sup f_j(x), \inf f_j(x), f_i + f_j, f_i f_j, \limsup f_j(x), \liminf f_j(x), \max(f_i, f_j), \min(f_i, f_j)$ *are all measurable.*
2. *Define $f^+(x) := \max(f(x), 0)$, $f^-(x) := \min(-f(x), 0)$, then f is measurable $\Leftrightarrow$ f^+, f^- are both measurable.*

Theorem 12. *For any $f : X \to \mathbb{C}$, define sgn$(f(z)) = f(x)/|f(x)|, 0$ when $|f(x)| = 0$, then f is measurable $\Leftrightarrow$ $|f|$, sgn(f) are both measurable.*

Now we think about approximating measurable functions, it turns out there is a good notation that "integrate" well:

Definition 9. A function ϕ is **simple** if it takes only finitely many values. In this case, we can write $\phi = \sum_{j=1}^{n} a_j \chi_{E_j}$.

Theorem 13. *Let $(X, \mathcal{M})$ be a measurable space. If $f : X \to [0, \infty]$ is measurable, there is a sequence $\{\phi_n\}$ of simple functions such that*

- $0 \le \phi_1 \le \phi_2 \le \cdots \le f$,
- $\phi_n \to f$ *pointwise, and*
- $\phi_n \to f$ *uniformly on any set on which f is bounded.*

The same theorem holds for $f : X \to \mathbb{C}$, with actual functions replaced by absolute value in the first statement.

We also could allow some "mistakes" in some points:

Theorem 14. *μ is complete* $\iff$ $\begin{cases} f \text{ is measurable, } f = g \ \mu\text{-a.e.}, \Rightarrow g \text{ is measurable.} \\ f_n \text{ is measurable, } f_n \to f \ \mu\text{-a.e.}, \Rightarrow g \text{ is measurable.} \end{cases}$

4 The Integral and Convergence Theorems

Throughout this section, we fix a measure space $(X, \mathcal{M}, \mu)$. (e.g. the definition of L^+ should be $L^+(X, \mathcal{M}, \mu)$)

Definition 10. If $\phi = \sum_{j=1}^{n} a_j \chi_{E_j}$ is simple, **The integral of** ϕ is defined as $\int \phi \, d\mu := \sum_{i=1}^{n} a_i \mu(E_i)$, also define

$$L^+ := \{f | f : X \to [0, \infty] \text{ is measurable}\}$$

, and we can define $\int f \, d\mu = \sup \left\{ \int \phi \, d\mu : 0 \le \phi \le f, \phi \text{ simple} \right\}$ for any $f \in L^+$.

Definition 11. A function $f : X \to \overline{\mathbb{R}}$ (resp. $\mathbb{C}$) is **integrable** if

1. For $\mathbb{R}$, $\int |f| = \int f^+ + \int f^- < \infty$
2. For $\mathbb{C}$, Ref, Imf are both measurable (rely on definition of integrable in $\overline{\mathbb{R}}$).

In this case, we define

$$L^1(X,\mathbb{R}) = \{f | f : X \to \mathbb{R} \text{ is integrable}\}.$$

We also define $\int f := \int f^+ - \int f^-$ (resp. $:= \int \mathrm{Re} f + i \int \mathrm{Im} f$) for any $f \in L^1$.

Theorem 15 (Properties of Integral).

- *$\int$ is a linear function: for all $f \in L^1, a\int f + b\int g = \int(af + bg)$.*
- *(Triangle Inequality.) If $f \in L^+$, then $f \le g \Rightarrow \int f \le \int g$. If $f \in L^1$, then $|\int f| \le \int |f|$.*
- *If $f \in L^+$, then define $\lambda(E) = \int_E f\, d\mu := \int f\chi_E$, then λ is a measure (this have good synergy with continuity from above/below! - in particular it is used to prove MCT.)*

The reason of developing the whole Lebesgue measure, one of them is that integral generalizes very well, in particular, existence and convergence of integral are very well-behaved:

Theorem 16 (MCT, Monotone Convergence Theorem). *Let $\{f_n\}$ be a sequence of measurable function in L^+, now if*

- *$\{f_n\}$ is monotone, i.e. $0 \le f_n(x) \le f_{n+1}(x)$ for all $x \in X$*
- *$f_n \to f$ pointwise, i.e. $\lim_{n\to\infty} f_n(x) = f(x)$ for all $x \in X$*

then $f \in L^+$ and $\int f = \lim_{n\to\infty} \int f_n$.

Theorem 17 (DCT, Dominated Convergence Theorem). *Let $\{f_n\}$ be a sequence of measurable function in L^1, now if*

- *There is a dominated function over $\{f_n\}$, i.e. there exists $g \in L^1$ such that $|f_n(x)| \le g(x)$ for all $x \in X$, and for all $n \in \mathbb{N}$.*
- *$f_n \to f$ pointwise, i.e. $\lim_{n\to\infty} f_n(x) = f(x)$ for all $x \in X$*

then $f \in L^1$ and $\int f = \lim_{n\to\infty} \int f_n$.

Now note that the blue for all $x \in X$, this can be strengthen to μ-a.e.

Theorem 18 (Consequneces of MCT). *If $\{f_n\} \in L^+$, then*

- *(Chebyshev's Inequality.) $\dfrac{\int f}{a} \le \mu(f^{-1}([a,\infty])$.*
- *(Fatou's Lemma.) $\int(\liminf f_n) \le \liminf \int f_n$*

Theorem 19 (Exchange Integrals and Sums). *If $\{f_n\} \in L^+$, then $\int \sum f_n = \sum \int f_n$.*
If $\{f_n\} \in L^1$ with $\sum \int |f_n| < \infty$, then $\int \sum f_n = \sum \int f_n$

Theorem 20 (Integral being zero). *If $f \in L^+$, then $\int f = 0 \Leftrightarrow f \equiv 0$ a.e.*
If $f, g \in L^1$, then for all $E \in \mathcal{M}$,

$$\int_E f = \int_E g \Leftrightarrow \int |f - g| = 0 \Leftrightarrow f = g \text{ a.e.}$$

Theorem 21 (Approximation of Integrals by Functions). *If $f \in L^+$, then there is always an integrable simple function ϕ such that $\int |f - \phi|\, d\mu < \varepsilon$. If μ is a Lebesgue-Stieltjes Meausre μ_G, then we can further conclude the existence ϕ as continuous.*

Theorem 22 (Interchange Differentiation and Integral). *Suppose $f : X \times [a,b] \to \mathbb{C}$ and $f(\cdot, t) : X \to \mathbb{C}$ is integrable for each $t \in [a,b]$. Let $F(t) = \int_X f(x,t) d\mu(x)$.*

- *Suppose there exists $g \in L^1(\mu)$ such that $|f(x,t)| \le g(x)$ for all x, t. If $\lim_{t\to t_0} f(x,t) = f(x,t_0)$ for every x, then $\lim_{t\to t_0} F(t) = F(t_0)$; in particular, if $f(x,\cdot)$ is continuous for each x, then F is continuous.*
- *Suppose $\partial f/\partial t$ exists and there is a $g \in L^1(\mu)$ such that $|(\partial f/\partial t)(x,t)| \le g(x)$ for all x, t. Then F is differentiable and $F'(t) = \int(\partial f/\partial t)(x,t) d\mu(x)$*

Theorem 23 (Riemann Integration). *Let f be a bounded real-valued function on $[a,b]$.*

- *f is Riemann integrable $\Rightarrow$ f is Lebesgue measurable.*
- *f is Riemann integrable $\Leftrightarrow \{x \in [a,b] : f$ is discontinuous at $x\}$ has Lebesgue measure zero.*

5 More on Convergence

We compare 5 types of convergence: (a.e.) pointwise, uniform, and the following three:

Definition 12 (Modes of Convergence)**.** Given a sequence of (complex-valued) functions $\{f_n\}$:

- $\{f_n\}$ **converges in** L^1 **to** f if the real sequence $\int |f_n - f| \to 0$
- $\{f_n\}$ **is Cauchy in measure** if for every $\varepsilon > 0$, $\mu(\{x : |f_n(x) - f_m(x)| \geq \varepsilon\}) \to 0$ as $m, n \to \infty$,
- $\{f_n\}$ **converges in measure to** f if for every $\varepsilon > 0$, $\mu(\{x : |f_n(x) - f(x)| \geq \varepsilon\}) \to 0$ as $n \to \infty$.
- $\{f_n\}$ is **almost uniform converges to** f if for all $\varepsilon > 0$, there exists $E \subset X$ such that $\mu(E) < \varepsilon$ and f_n uniformly converges to f on E^c.

We start collecting implications and present counter examples:

- Convergences in $L^1 \Rightarrow$ Convergence in Measure. Counterexample for converse: $f_n = n\chi_{(0,1/n)}$
- Cauchy in measure $\Leftrightarrow$ Convergence in measure.
  - In fact if $\mu(X)$ is finite, we can build a metric space $(L^1/\sim, \rho)$ where convergence is exactly convergence in measure. ($\sim$ is the relation identifying a.e. functions) $\Rightarrow$ convergence of measure is unique up to a.e.
- Convergence in measure to $f \Rightarrow$ There exist a subsequence $\{f_{n_j}\}$ such that $f_{n_j} \to f$ pointwise a.e.
- Given $\mu(X) < \infty$, then $f_n \to f$ a.e. pointwise $\Leftrightarrow$ f_n almost uniform converge to f.
  - Note that right-to-left always holds even with $\mu(X) = \infty$.

  Left-to-right Doesn't hold when $\mu(X) = \infty$: $f_n = \chi_{(n,n+1)}$ (Not converging in measure), $f_n = \chi_{[H_n, H_{n+1})}$, where H_n is the nth harmonic number. (converging in measure)
- Almost uniform Convergence $\Rightarrow$ convergence in measure. Counterexample: Typewriter sequence, note that this doesn't even converge a.e.

6 Product σ-algebras and Product Measures

Definition 13. Given $f_\alpha : X \to Y_\alpha$, where $\{Y_\alpha, \mathcal{N}_\alpha\}_{\alpha \in \mathcal{A}}$ is a collection of measure spaces, indexed by $\alpha \in \mathcal{A}$. then let

$$\mathcal{E} = \left\{ f_\alpha^{-1}(E_\alpha) : E_\alpha \in \mathcal{N}_\alpha, \alpha \in \mathcal{A} \right\}$$

, (Then by definition all the f_α are $(\mathcal{M}, \mathcal{N}_\alpha)$ measurable.) $\sigma(\mathcal{E})$ is called the σ**-algebra generated by** $\{f_\alpha\}_{\alpha \in \mathcal{A}}$.

Given a collection of measure space $\{Y_\alpha, \mathcal{N}_\alpha\}_{\alpha \in \mathcal{A}}$, we can always define the Cartesian product $Y^* := \prod_{a \in \alpha} Y_\alpha$

Definition 14. The product σ**-algebra of** $Y_\alpha := \bigotimes_{\alpha \in \mathcal{A}} Y_\alpha$ is the σ-algebra generated by $\{\pi_\alpha\}_{\alpha \in \mathcal{A}}$, the projection maps $Y^* \to Y_\alpha$.

Theorem 24. *The product is a product in the categorical sense: given any measurable function $f : X \to Y^*$: we must have f is measurable $\Leftrightarrow$ $\pi_\alpha \circ f$ is measurable for all $\alpha \in \mathcal{A}$*

Theorem 25. $\mathbb{R}^n = \bigotimes_1^n \mathbb{R}$ *where $\mathbb{R}$ is equipped with the σ-algebra.*

Now, given two measure spaces $(X, \mathcal{M}, \mu), (Y, \mathcal{N}, \nu)$, we are ready to define the product measure space.

Definition 15. A **(measurable) rectangle** is the Cartesian product of measurable sets in $\mathcal{M}, \mathcal{N}$ respectively.

This is an algebra of sets in $X \times Y$. So we can define

Definition 16. The **premeasure** on the algebra of rectangle is:

$$\mu \times \nu(A \times B) := \mu(A)\nu(B).$$

After checking that it satisfies countable additivity, by theorem 4:

- For any μ, ν there is an extension to $\mathcal{M} \otimes \mathcal{N}$ by the outer measure.
- Moreover, if both μ, ν are σ-finite, then such extension is unique.

We call this the **product measure** and denote it by $\mu \times \nu$ as above.
Now we want to focus our proof in the following way: conclude for finite measure space, conclude on all rectangles, and then conclude the whole statement (usually σ-finite measure space). The definition of Monotone classes can do this:

Definition 17. $\mathcal{C}$ is a **monotone class** if it is closed under countable increasing union and countable decreasing intersections. (Intuitively, think about, collecting increasing/decreasing circles)

Given a collection of sets, there is the smallest monotone class generated by it, and

Theorem 26. *If $\mathcal{A}$ is an algebra, the smallest monotone class generated by $\mathcal{A}$ is the σ-algebra generated by $\mathcal{A}$.*

Now we can establish some theorem, especially on how to calculate the product measure $\mu \times \nu(E)$ and integrating functions on $X \times Y$.

Definition 18 (x and y-section)**.** . The x_0**-section**

- of a set E is denoted by $E_{x_0} := \{y \in Y : (x_0, y) \in E\}$.
- of a function f is denoted by $f_{x_0}(y) := f(x_0, y) = f(\cdot, y)$, note that it is a function $Y \to \mathbb{R}$.

The y**-section** is defined similarly, with superscripts being used instead.

Theorem 27 (Evaluating product measure)**.** *Given $(X, \mathcal{M}, \mu), (Y, \mathcal{N}, \nu)$ are σ-finite measure spaces and $E \in \mathcal{M} \otimes \mathcal{N}$, then $x \to \nu(E^x), y \to \mu(E^y)$ are measurable on X, Y (i.e. L^+) respectively, and*

$$\mu \times \nu(E) = \int \nu(E_x)\, d\mu(x) = \int \mu(E^y)\, d\nu(y)$$

Theorem 28 (Fubini-Tonelli)**.** *Given $(X, \mathcal{M}, \mu), (Y, \mathcal{N}, \nu)$ are σ-finite measure spaces then if f is $L^1(X \times Y)$ or $L^+(X \times Y)$, then*

$$\int f\, d(\mu \times \nu) = \int f\, d\nu\, d\mu = \int f\, d\mu\, d\nu$$

holds. Moreover, define

$$g(y) = \int f_x(y)\, d\nu \qquad h(x) = \int f^y(x)\, d\mu$$

then if $f \in L^+(X \times Y)$ (resp. $f \in L^1(X \times Y)$) then $g, h \in L^+(X), L^+(Y)$ (resp. $\in L^1(X), L^1(Y)$).

7 Signed Measures and Decompositions

Definition 19. A **signed measure** on $(X, \mathcal{M})$ is a function $\nu : \mathcal{M} \to [-\infty, \infty]$ such that

- $\nu(\varnothing) = 0$, ν assumes at most one of the values $\pm\infty$, and
- (Countable additivity.) If $\{E_j\}$ is a sequence of disjoint sets in $\mathcal{M}$, then $\nu\left(\cup_1^\infty E_j\right) = \Sigma_1^\infty \nu\left(E_j\right)$, where the latter series converges absolutely if $\nu\left(\cup_1^\infty E_j\right)$ is finite.

Definition 20. Given a measure μ, a function f is called **extended μ-integrable** if any one of $\int f^+\, d\mu$, $\int f^-\, d\mu$ is finite. Given such function, the function $\nu(E) = \int_E f\, d\mu$ is a measure, and we denote this by $d\nu = f\, d\mu$ or we say **"the measure** $f\, d\mu$**"** in place of ν.

The problem of looking at signed measure is that it loses sub-additivity. However, it turns out the issue is not bad, because we can decompose it to measures. The idea is: it turns out that adding two measures and integrating with respect to a function f can give us all signed measures. We can use decomposition to see that.

Definition 21. Fix a signed measure ν, a set E is **positive** (resp. **negative, null**) , if $\nu(F) \geq 0 (\leq 0, = 0)$ for all $F \subset E$.

Definition 22. Two signed measures μ and ν are **mutually singular**, written $\mu \perp \nu$, if there exist $E, F \in \mathcal{A}$ such that

- $E \cap F = \varnothing, E \cup F = X$, and
- E is null for μ, and F is null for ν.

Informally speaking, mutual singularity means that μ and ν "live on disjoint sets."

Definition 23. Let μ be a measure and ν a signed measure, then we say ν is **absolutely continuous to** μ, written $\nu \ll \mu$, if $\mu(E) = 0 \Rightarrow \nu(E) = 0$. The name comes from the following equivalent definition:

Theorem 29. *$\nu \ll \mu \Leftrightarrow$ for all $\varepsilon > 0$, there exists $\delta > 0$ such that $\mu(E) < \delta \Rightarrow |\nu(E)| < \varepsilon$.*

Theorem 30 (Hahn Decomposition)**.** *If ν is a signed measure on $(X, \mathcal{M})$, there exist a positive set P and a negative set N for ν such that $P \cup N = X$ and $P \cap N = \varnothing$. If P', N' is another such pair, then $P \Delta P'$ $(= N \Delta N')$ is null for ν.*

So we have a decomposition of sets, from here

Theorem 31 (Jordan Decomposition)**.** *If ν is a signed measure, there exist **unique** positive measures $\nu^+$and ν^-, defined by*

$$\nu^+(E) = \nu(E \cap P), \nu^-(E) = -\nu(E \cap N).$$

where $X = P \cup N$ be a Hahn decomposition for ν, it follows that $\nu = \nu^+ - \nu^-$and $\nu^+ \perp \nu^-$.

Uniqueness will come from the null condition on Hahn decomposition. Now we are good to define more definitions that synergize with signed measures.

Definition 24. Given a signed measure ν and its Jordan decomposition ν^+, ν^-:

- The **total variation** of a signed measure is the measure $|\nu| := \nu^+ + \nu^-$, and ν is **finite** or **σ-finite** if $|\nu|$ is.
- Integrate f with respect to ν is defined by $L^1(\nu) = L^1(\nu^+) \cap L^1(\nu^-)$ and $\int f \, d\nu := \int f \, d\nu^+ - \int f \, d\nu^-$.

Theorem 32. *$\nu \ll \mu \Leftrightarrow |\nu| \ll \mu \Leftrightarrow \nu^+ \ll \mu$ and $\nu^- \ll \mu$, in particular, the measure $d\nu = f d\mu \ll \mu$*

Now we can fully decompose the measure using the following lemma.

Lemma. *If ν, μ are finite measures, then either*

- *$\nu \perp \mu$, or*
- *For all $\varepsilon > 0$, There exists measurable set $\mu(E) > 0$ such that E is a positive set of the measure $\nu - \varepsilon\mu$.*

Proof. Consider the Hahn decomposition of $\nu - \dfrac{\mu}{n}$. □

Theorem 33 (Lebesgue-Radon-Nikodym)**.** *Let ν be a σ-finite signed measure and μ a σ-finite measure. Then there exists unique σ-finite signed measure λ, ρ such that:*

- *$\lambda \perp \mu$, $\rho \ll \mu$, and $\nu = \lambda + \rho$.*
- *There is a unique up to μ-a.e., extended μ-integrable function f such that $d\rho = f \, d\mu$.*

In particular, the case where $\nu \ll \mu$, then $\lambda = 0, \rho = \nu$, this is referred to as the Radon-Nikodym Theorem.

Definition 25. The function f in Radon-Nikodym theorem is referred as the **Radon-Nikodym derivative**, and is written as $\dfrac{d\nu}{d\mu}$ (note: we have $d\nu = \dfrac{d\nu}{d\mu} d\mu$ as the statement of Radon-Nikodym theorem).

Theorem 34 (Chain Rule)**.** *If ν is a σ-finite signed measure, μ, λ are σ-finite measures, $\nu \ll \mu$, $\mu \ll \lambda$, then $\nu \ll \lambda$, $\dfrac{d\nu}{d\lambda} = \dfrac{d\nu}{d\mu}\dfrac{d\mu}{d\lambda}$. Also ,If $g \in L^1(\nu)$, then $g \cdot \left(\dfrac{d\nu}{d\mu}\right) \in L^1(\mu)$ and $\displaystyle\int g \, d\nu = \int g \cdot \left(\frac{d\nu}{d\mu}\right) d\mu$.*

Definition 26. Writing $\nu = \lambda + \rho$ is called the **Lesbesgue Decomposition of ν with respect to μ.**

8 Differentiation in $(\mathbb{R}^n, m)$

We want to recover the differentiation process by first principles.

Definition 27. A **locally integrable function** $f \in L^1_{loc}$ is a function such that $\int_K |f|\, dm < \infty$ for all bounded $K \in \mathbb{R}^n$.
The **moving average** of f is $A_r f(x) := \dfrac{\int_{B(r,x)} f(y)dy}{m(B(r,x))}$.
The **Hardy-Littlewood maximal function** of a function f is defined as $Hf(x) := \sup_{r>0} A_r|f|(x)$.

The reason why we choose the balls $B(r,x)$ radius r, centered at x is because of the follow theorem:

Theorem 35. *Given a collection of open balls and union them to get an open set U, then if $c < m(U)$, there exists disjoint open balls $B_1 ... B_k$ such that $\sum_j m(B_j) > \dfrac{c}{3^n}$*

Theorem 36 (Maximal Theorem)**.** *There is a constant $C > 0$ such that for all $f \in L^1, \alpha > 0$,*

$$m(\{x : Hf(X) > \alpha\}) \leq \frac{C}{\alpha} \int |f|\, dx$$

Differentiation actually makes sense a.e., but even better, the Lebesgue Set:

Definition 28. The **Lebesgue Set** of a function f is

$$L_f := \left\{ x : \lim_{r\to 0} \frac{\int_{B(r,x)} |f(y) - f(x)|\, dy}{m(B(r,x))} = 0 \right\}$$

Theorem 37. *Lebesgue Set is full for L^1_{loc} functions. That is, $m(L_f^c) = 0$.*

Theorem 38 (Lebesgue Differentiation Theorem)**.** *Given $f \in L^1_{loc}, x \in L_f$, we have $f(x) = \lim_{r\to 0} \dfrac{\int_{B(r,x)} f(y)\, dy}{m(B(r,x))}$*

This almost gives an immediate application by synergizing with the Lebesgue decomposition. But the above theorem only holds for L^1_{loc} function, but not all L^1 functions. So we need to restrict ourselves to nicer measures:

Definition 29. A measureν is **regular** if it is:

- (Inner Regular.) $\nu(K) < \infty$ for all compact K, and
- (Outer Regular.) $\nu(E) = \inf\{\nu(U) : E \subset U, U \text{ open}\}$ for all $E \in \mathcal{B}_{\mathbb{R}^n}$.

Theorem 39. *If ν is regular then if $\nu = \lambda + \rho = \lambda + f\, dm$ as the Lebesgue decomposition, then*

$$\frac{d\nu}{dm} := f(x) = \lim_{r\to 0} \frac{\nu(B(r,x))}{m(B(r,x))}$$

for m-a.e x.

We don't need to consider just $B(r,x)$, but any collection of sets that satisfies theorem 35:

Definition 30. A family $\{E_r\}_{r>0}$of Borel subsets of $\mathbb{R}^n$ is said to **shrink nicely** to $x \in \mathbb{R}^n$ if

- $E_r \subset B(r,x)$ for each r; and
- there is a constant $\alpha > 0$, independent of r, such that $m(E_r) > \alpha m(B(r,x))$.

From definition 28 onwards, all conclusion still holds if balls are replaced by general sets that shrink nicely.

9 Functions from Measures, FTC revisited

Recall in section 2, we define (Theorem 7) the Lebesgue-Stieltjes Measure associated to an increasing, right-continuous F as μ_F, now we can synergize $\frac{d\mu_F}{dm}$ is exactly the derivative F':

Theorem 40. *Let $F : \mathbb{R} \to \mathbb{R}$ be increasing, and let $G(x) = F(x+)$. Then*

- *The set of points at which F is discontinuous is countable.*
- *F and G are differentiable a.e., and $F' = G'$ a.e.*

Definition 31.

$$T_F(x) = \sup\left\{\sum_1^n |F(x_j) - F(x_{j-1})| : n \in \mathbb{N}, -\infty < x_0 < \cdots < x_n = x\right\}$$

T_F is called the **total variation function** of F. If $\lim_{n\to\infty} T_f(x) < \infty$, then we write $T \in BV$ and say T is of **bounded variation**. Also, define

$$NBV = \{F \in BV : F \text{ is right continuous and } \lim_{x\to-\infty} F(x) = 0\},$$

note that all $f \in BV$ have a corresponding NBV function g: $g(x) := f(x+) - \lim_{x\to-\infty} f(x)$, and $g' = f'$ a.e.

Theorem 41. *The first theorem in this section also holds if $F \in BV$. Moreover, $F \in BV$ implies F has only countably many discontinuity.*

Theorem 42 (Jordan Decomposition)**.** *$F \in BV \Leftrightarrow F$ is a difference of two bounded, increasing function. In this case, we can write $F = \frac{1}{2}(T_F + F) - \frac{1}{2}(T_F - F)$, that satisfies the requirement above and is referred as **Jordan Decomposition** of F.*

The μ_F association can be done in more generality. A **Borel measure** is a measure define on all Borel sets.

Theorem 43.

- *μ is a complex Borel measure, then $\mu = \mu_F$ for some $F \in NBV$.*
- *If $F \in NBV$, then there is a unique complex borel measure μ_F such that $\mu_F((-\infty, x]) = F(x)$.*

Theorem 44 (Properties of μ_F)**.**

- $F \in NBV \Rightarrow F' \in L^1(\mu)$.
- $\mu_F \perp m \Leftrightarrow F' = 0$ *a.e.*
- $\mu_F \ll m \Leftrightarrow F(x) = \int_{-\infty}^x F'(t)\,dt \Leftrightarrow$ *if for every $\varepsilon > 0$ there exists $\delta > 0$ such that for any finite set of disjoint intervals $(a_1, b_1), \ldots, (a_N, b_N)$,*

$$\sum_1^N (b_j - a_j) < \delta \Rightarrow \sum_1^N |F(b_j) - F(a_j)| < \varepsilon.$$

Note the last condition on the last bullet point only depends on the function F.

Definition 32. We simply refer to that being F is **absolutely continuous**, we say F is absolutely continuous on $[a, b]$ if the finite set of disjoint intervals are in $[a, b]$. Note that this implies $F \in BV([a, b])$

With all of these set up we finally get:

Theorem 45 (FTC for Lebesgue Integrals)**.** *If $-\infty < a < b < \infty$ and $F : [a, b] \to \mathbb{C}$, the following are equivalent:*

- *F is absolutely continuous on $[a, b]$.*
- *$F(x) - F(a) = \int_a^x f(t)dt$ for some $f \in L^1([a, b], m)$.*
- *F is differentiable a.e. on $[a, b]$, $F' \in L^1([a, b], m)$, and $F(x) - F(a) = \int_a^x F'(t)dt$.*

Application of Fubini (calculating what μ_F, μ_G is, we also get):

Theorem 46 (Chain Rule)**.** *If F and G are in NBV and at least one of them is continuous, then for $-\infty < a < b < \infty$:*

$$\int_{(a,b]} F d\mu_G + \int_{(a,b]} G d\mu_F = F(b)G(b) - F(a)G(a).$$